

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801A - CHARACTERIZATION OF CONSTRUCTION MATERIALS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

- Q1. Objective type questions. (Compulsory Question) 12
- 1 The microstructure of a construction material primarily influences its:
A) Color and appearance
B) Mechanical and durability properties
C) Market cost
D) Availability
 - 2 Cement paste, aggregates, and ITZ together describe concrete at the:
A) Atomic scale
B) Nano scale
C) Microstructural scale
D) Molecular scale
 - 3 Calorimetry in cement chemistry is mainly used to study:
A) Particle size distribution
B) Heat evolution during hydration
C) Chemical composition
D) Porosity distribution
 - 4 X-ray diffraction is primarily used to identify:
A) Pore size distribution
B) Moisture content
C) Surface texture
D) Crystalline phases in materials
 - 5 Broad XRD peaks generally indicate:
A) Large crystals
B) High crystallinity
C) Poor crystallinity or nano-sized crystals
D) High density

- 6 Differential Thermal Analysis (DTA) measures:
 - A) Heat flow difference between sample and reference
 - B) Mass loss only
 - C) Density variation
 - D) Electrical resistance
- 7 Higher surface area of cement generally leads to:
 - A) Lower rate of hydration
 - B) Faster hydration and higher early strength
 - C) Reduced water demand
 - D) Lower heat evolution
- 8 Optical microscopy is mainly used to study:
 - A) Atomic lattice
 - B) Microstructure at micrometer scale
 - C) Chemical bonds
 - D) Electrical properties
- 9 SEM provides information mainly about:
 - A) Surface morphology and microstructure
 - B) Chemical composition only
 - C) Internal crystal lattice
 - D) Heat evolution
- 10 Image analysis in construction materials is mainly used to quantify:
 - A) Chemical composition
 - B) Heat of hydration
 - C) Pore size, shape, and distribution
 - D) Density
- 11 Spectroscopic techniques are primarily based on:
 - A) Mechanical interaction
 - B) Fluid flow
 - C) Heat transfer
 - D) Interaction of electromagnetic radiation with matter
- 12 Mercury intrusion porosimetry is used to determine:
 - A) Pore size distribution
 - B) Chemical composition
 - C) Heat of hydration
 - D) Elastic modulus

Q2. Solve the following.

- A) Explain the structure of construction materials and discuss the importance of material characterization. 6
- B) Describe the principle of calorimetry and explain its application in construction materials. 6

Q3. Solve the following.

- A) Explain the principle of X-ray diffraction. 6
- B) Describe thermal analysis techniques. 6

Q4. Solve Any Two of the following.

- A) Explain surface area measurement techniques. 6
- B) Describe optical microscopy and its applications in construction material studies. 6

- C) Explain scanning electron microscopy (SEM). 6
- Q5. Solve Any Two of the following.
- A) Explain image analysis techniques and their role in material characterization. 6
 - B) Describe spectroscopic techniques used in construction materials analysis. 6
 - C) Explain the working principle of mercury intrusion porosimetry. 6
- Q6. Solve Any Two of the following.
- A) Explain impedance analysis and its significance in evaluating material properties. 6
 - B) Describe ultrasonic testing methods and their applications in construction materials. 6
 - C) Compare destructive and non-destructive material characterization techniques. 6

*** End ***